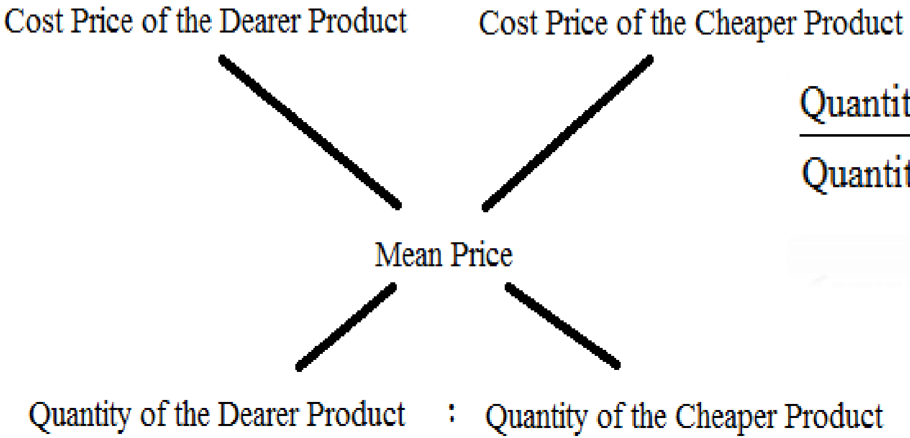
MIXTURE AND ALLIGATION

ALLIGATION Rule that enables us find the ratio in which two or more ingredients at the given price must be mixed to produce a mixture of a desired price.

MEAN PRICE Cost price of a unit quantity of the mixture.

RULE OF MIXING



$$\frac{\text{Quantity of the Cheaper Ingredient}}{\text{Quantity of the Dearer Ingredient}} = \frac{CP_{Dearer} - CP_{Mean}}{CP_{Mean} - CP_{Cheaper}}$$

PRACTICE QUESTIONS

In what ratio must rice at Rs. 9.30 per kg be mixed with rice at Rs. 10.80 per kg so that the mixture be worth Rs. 10 per kg?

A. 8:7 **B. 7:8** **C. 10:9** **D.9:10**

By the rule of alligation, we have:

C.P. of 1 kg rice of 1st kind C.P. of 1 kg rice of 2nd kind

930 **1080**

Mean Price

1000

80 **70**

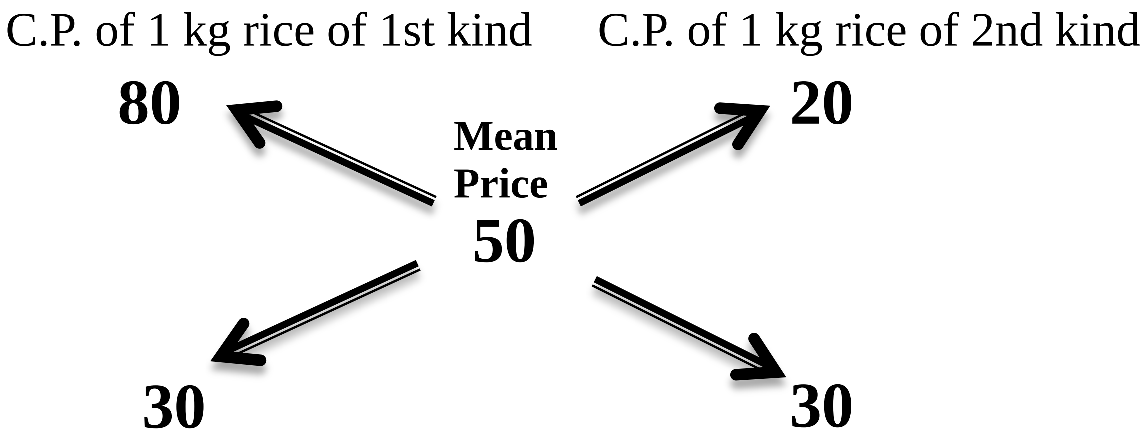
Hence, Required ratio = 80 : 70 = 8 : 7.

PRACTICE QUESTIONS

If we purchase two varieties of rice worth Rs. 80 and Rs. 20 per kilogram, to make a mixture worth Rs. 50 per kilogram, then the ratio in which the ingredients are mixed is:

A. 1:1 **B. 1:2** **C. 2:1** **D. 2:3**

By the rule of alligation, we have:



Hence, Required ratio = 30 : 30 = 1 : 1.

PRACTICE QUESTIONS

The cost of Type 1 rice is Rs. 15 per kg and Type 2 rice is Rs. 20 per kg. If both Type 1 and Type 2 are mixed in the ratio of 2 : 3, then the price per kg of the mixed variety of rice is:

A. Rs 18 **B. Rs 17** **C. Rs 19** **D. Rs 17.50**

Let the price of the mixed variety be Rs. *x* per kg.

By rule of alligation, we have:

C.P. of 1 kg rice of 1st kind C.P. of 1 kg rice of 2nd kind

15 **20**

Mean Price

x

20-x ***x-15***

According to question

$$\frac{20-x}{x-15}=\frac{2}{3}$$

$$60-3x=2x-30$$

$$-3x-2x=-30-60$$

$$-5x=-90$$

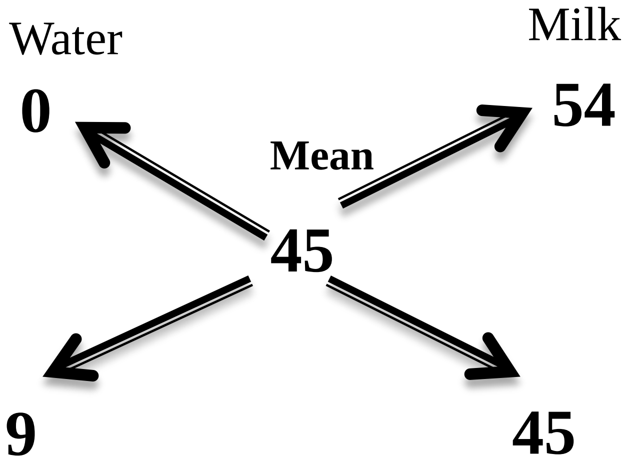
$$x=18$$

PRACTICE QUESTIONS

A mixture of certain quantity of milk with 8 L of water is worth 45 paisa per litre. If pure milk be worth 54 paisa per litre, how much milk is three in the mixture?

(A) 40 L　　(B) 35 L　　(C) 25 L　　(D) 45 L

By rule of alligation, we have:



Ratio is　9:45

or 1:5

So quantity of milk in mixture

$=5\times 8$

$=40$ Lit

Ans A

PRACTICE QUESTIONS

A trader has 50 kg of pulse, part of which he sells at 8% profit and rest at 18% profit. He gains 14% on the whole. What is the quantity sold at 18% profit?

(A) 30 kg **(B) 35 kg** **(C) 40 kg** **(D) 60 kg**

By rule of alligation, we have:

Part I Part II

8% **18%**

Mean

14%

4% **6%**

Ratio is 4:6

or 2:3

So quantity sold at 18% profit

$= \frac{3}{5} \times 50$

=30 Kg

Ans A

REPLACEMENT OF PART OF SOLUTION

- Suppose a container contains *x* units of liquid from which *y* units are taken out and replaced by water.
- After *n* operations, the quantity of pure liquid $= x(1 - y/x)^n$ units
- If each time different quantities is taken out, then

  the quantity of pure liquid $= x\left(1 - \frac{y_1}{x}\right)\left(1 - \frac{y_2}{x}\right)\left(1 - \frac{y_3}{x}\right)$ units

From a cask of milk containing 30 liters, 6 liters are drawn out and the cask is filled up with water. If the same process is repeated a second, then a third time, what will be the number of liters of milk left in the cask?

Amount of milk left $= 30(1\text{-}6/30)^3$

$$= 30 \times \left(1 - \frac{1}{5}\right)^3$$

$$= 30 \times \frac{4}{5} \times \frac{4}{5} \times \frac{4}{5} \quad = 15.36\, Lit$$

PRACTICE QUESTIONS

A container contains 40 liters of milk. From this container 4 liters of milk was taken out and replaced by water. This process was repeated further two times. How much milk is now contained by the container?

Amount of milk left $= 40(1-4/40)^3$

$= 29.16$ Liters

PRACTICE QUESTIONS

8 litres are drawn from a cask full of wine and is then filled with water. This operation is performed three more times. The ratio of the quantity of wine now left in cask to that of water is 16 : 65. How much wine did the cask hold originally?

Let the quantity of the wine in the cask originally be x liter.

Then, quantity of wine left in cask after 4 operations $= x(1–8/\ x)^4$ litres

$$x(1 - (8/x))^4 \;= 16 / 81$$

$$x = 24 \text{ liter}$$

PRACTICE QUESTIONS

A vessel is filled with liquid, 3 parts of which are water and 5 parts syrup. How much of the mixture must be drawn off and replaced with water so that the mixture may be half water and half syrup?

A.1/3 **B.1/4** **C.1/5** **D.1/7**

Suppose the vessel initially contains 8 litres of liquid.

Let x litres of this liquid be replaced with water.

Quantity of water in new mixture $= 3 - \frac{3x}{8} + x$

Quantity of syrup in new mixture $= 5 - \frac{5x}{8}$

According to question

$$3 - \frac{3x}{8} + x = 5 - \frac{5x}{8}$$

$$\Longrightarrow 5x + 24 = 40 - 5x$$

$$\Longrightarrow 10x = 16$$

$$\Longrightarrow x = \frac{8}{5}$$

So, part of the mixture replaced

$$= \frac{8}{5} \times \frac{1}{8} \qquad = \frac{1}{5}$$

Tea worth Rs. 126 per kg and Rs. 135 per kg are mixed with a third variety in the ratio 1 : 1 : 2. If the mixture is worth Rs. 153 per kg, the price of the third variety per kg will be:

A. Rs. 169.50 B. Rs 170 C. Rs. 175.50 D. RS 180

Since first and second varieties are mixed in equal proportions.

So avg price =(126+135)/2=130.5

So, the mixture is formed by mixing two varieties, one at Rs. 130.50 per kg and the other at say, Rs. *x* per kg in the ratio 2 : 2, *i.e.*, 1 : 1. We have to find *x*.

By rule of alligation, we have:

Part I		Part II
130.5		**x**
	Mean 153	
X-153		**22.5**

Ratio is 1:1

$x - 153 = 22.5$

$x = 175.5$

Ans c

PRACTICE QUESTIONS

A container contains a mixture of two liquids P and Q in the ratio of 7:5. When 9 liters of mixture is taken out and replaced with Q, the ratio becomes 7:9. Find the quantity of liquid P in the container.

A. 10 L **B. 20 L** **C. 21 L** **D.25 L**

Let the quantity of mixture in the container be $12x$ liters

(i.e., $7x$ liters of liquid P and $5x$ liters of liquid Q).

Quantity of P in mixture left $=7x - \frac{7}{12} \times 9$ $=7x - \frac{21}{4}$

Quantity of Q in mixture left $=5x - \frac{5}{12} \times 9$ $=5x - \frac{15}{4}$

ATQ

$$\frac{7x - \frac{21}{4}}{5x - \frac{15}{4} + 9} = \frac{7}{9}$$

Solving, we get

$x = 3\ lit$

Quantity of P in container=21 lit

A milk vendor has 2 cans of milk. The first contains 25% water and the rest milk. The second contains 50% water. How much milk should he mix from each of the containers so as to get 12 litres of milk such that the ratio of water to milk is 3 : 5?

A. 4 litres, 8 litres B. 6 litres, 6 litres

C. 5 litres, 7 litres D. 7 litres, 5 litres

Let the cost of 1 litre milk be Re. 1

Milk in 1 litre mix. in 1st can $= \frac{3}{4} lit$

C.P. of 1 litre mix. in 1st can Rs. $\frac{3}{4}$

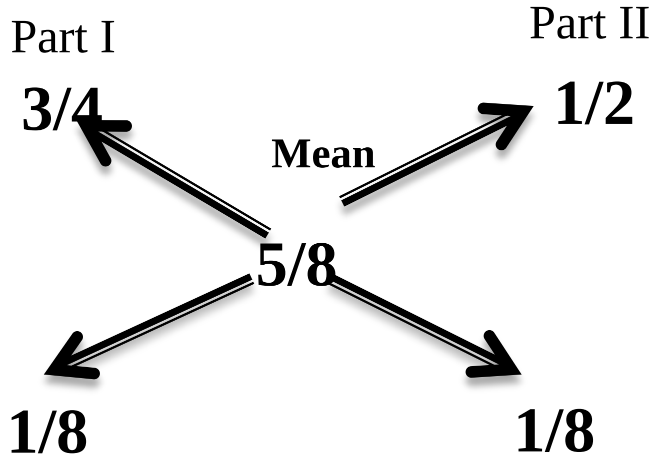
Milk in 1 litre mix. in 2nd can $= \frac{1}{2} lit$

C.P. of 1 litre mix. in 2nd can Rs. $\frac{1}{2}$

Milk in 1 litre of final mix. $= \frac{5}{8} lit$

C.P. of 1 litre mix in final . Rs. $\frac{5}{8}$

By rule of alligation, we have:



So ratio must be 1:1

So quantity taken from each can=6 lit

A dishonest milkman professes to sell his milk at cost price but he mixes it with water and thereby gains 25%. The percentage of water in the mixture is:

A. 4% **B. $6\frac{1}{4}$%** **C. 20%** **D. 25 %**

Let C.P. of 1 litre milk be Rs.100

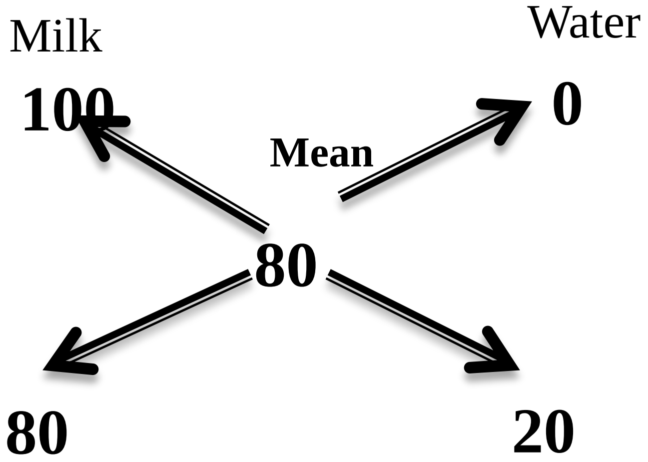
Gain =25%

So S.P. of 1 litre milk be Rs.125

But, ATQ SP=100

Means CP=80

By rule of alligation, we have:



Ratio is 80:20

or 4:1

So % water would be $\frac{1}{5} \times 100\%$

$= 20\%$

Ans C

How many kilogram of sugar costing Rs. 9 per kg must be mixed with 27 kg of sugar costing Rs. 7 per kg so that there may be a gain of 10% by selling the mixture at Rs. 9.24 per kg?

A. 36 kg B. 42 kg C. 54 kg D. 63 kg

S.P. of 1 kg of mixture = Rs. 9.24, Gain 10%.

C.P. of 1 kg of mixture = Rs. 8.4.

By rule of alligation, we have:

Part I   Part II

9   **7**

Mean

8.4

1.4   **0.6**

Ratio is 1.4:0.6

or 7:3

Now 3 part=27 kg

So 7 parts =63 kg

Ans D

1. The proportion in which a person mixes milk at ₹10.50 a liter with milk at ₹7.50 a liter so that the mixture may be worth ₹8.50 a liter?

A. 1:2 B. 4:5 C. Rs. 2:3 D. Rs. 1:3

2. The amount of water added to 9 liters of Alcohol worth ₹10 a liter so that the value of the mixture may be ₹7.50 a liter?

A. 5 liters B. 4 liters C. 3 liters D. 2 liters

3. In what proportion must Coffee Powder worth ₹80 and ₹100 per kg be mixed so as to gain 20 percent by selling the mixture at ₹98 per kg?

A. 11:1 B. 11:2 C. 1:11 D. 2:11

4. A person buys wine at ₹7 per liter and after mixing water sells the mixture at ₹8 per liter thereby making a profit of 25%. Then the proportion of water and wine in the mixture is?

A. 3:16 B. 3:32 C. 1:16 D. 2:11

5. A sum of ₹43.75 is made up of 125 coins which are of either 25 paise coins or 50 paise coins. How many coins are there of 50 paise?

A. ₹75 B. ₹50 C. ₹25 D. None of these

6. In a classroom of 142 children, boys have 1 rupee coins and girls have 25 paise and together possess a sum of ₹58. Then find the number of girls in the classroom.

A. 30 B. 84 C. 112 D. 118

7. In a mixture of milk and water in the ratio of 9:5, how much mixture should be withdrawn and water should be substituted in its place, so that in the resulting mixture, there may be half milk and half water?

A. 3/9 B. 2/9 C. 5/9 D. 1/9

8. One vessel contains a mixture of 3 parts milk and 1 part water, whereas the other vessel contains a mixture of 7 parts milk and 5 parts water. Compare the strength of the milk.

A. 7/9 B. 11/7 C. 9/7 D. 7/11

9. A merchant has 50 kgs of sugar, part of which he sells at 15% profit and the rest at 20% profit. He gains 17% on the whole. Find how much sugar is sold at 15% profit?

A. 30 kg B. 40 kg C. 70 kg D. 50 kg

10. In what ratio must a grocer mix two varieties of pulses costing ₹25 and ₹30 per kg respectively so as to get a mixture worth ₹28.5 per kg.

A. 4:5 B. 3:7 C. 2:5 D. 6:7

11. The cost of Type–I Wheat is ₹20 per kg and Type–II Wheat is ₹30 per kg. If both Type–I and Type–II are mixed in the ratio of 2:3, then price per kg of mixed variety of rice is

A. ₹24 B. ₹22 C. ₹28 D. ₹26

12. A mixture of a certain quantity of milk with 10 liters of water is worth ₹5.5 a liter. If pure milk be worth ₹10.5 a liter, how much milk is there in the mixture?

A. 10 liters B. 11 liters C. 12 liters D. 13 liters

13. 100 liters of mixture of milk and water contains 35% of water. How much water must be added to make 40% of water in new mixture?

A. 6.66 liters B. 7.33 liters C. 8.33 liters D. 9.66 liters

14. Solution A of 90% and solution B of 97% are mixed resulting in 21 liters of 94%. What is the quantity of first solution in resulting mixture?

A. 6 liters B. 7 liters C. 8 liters D. 9 liters

15. Two types of rice pricing ₹23 and ₹29 are mixed. The average price was ₹25. Another x% and y% of respective rice is mixed but the price was same i.e. ₹25 what will be the possible scenario?

A. $x > y$ B. $x < y$ C. $x = y$ D. cannot say

Answer question no. 16 to 18 using below data:
Type–I Rice = ₹15; Type–II Rice = ₹20; Type–III Rice = ₹25;

16. In what ratio Type–II and Type–III Rice should be mixed so as to get the average price of ₹24?
A. 4:1 B. 2:3 C. 1:4 D. 3:2

17. If Type–I and Type–III Rice are mixed and sold at ₹18. So, for 100 kg of Type–III Rice, how much kg of Type–I Rice should be mixed?
A. 200 kg B. 233 kg C. 266 kg D. 300 kg

18. What will be the per kg price of 20 kg of Type–I and 60 kg of Type–II Rice?
A. ₹18 B. ₹18.25 C. ₹18.50 D. ₹18.75

19. A 60 liter mixture of milk and water contains 10% water. How much water must be added to make water 20% in the mixture?

A. 8 liters B. 7.5 liters C. 7 liters D. 6.5 liters

20. A 20 liter mixture contains 30% alcohol and 70% water. If 5 liters of water is added to the mixture, what will be the percentage of alcohol in the new mixture ?

A. 22% B. 23% C. 24% D. 25%

21. A shopkeeper mixes 60 kg of sugar worth Rs. 30 per kg with 90 kg of sugar worth Rs. 40 per kg. At what rate he must sell the mixture to gain 20%?
A. Rs. 30 per kg B. Rs. 43.2 per kg C. Rs. 44.8 per kg D. Rs. 45 per kg

22.`700 ml of a mixture contains water and milk in the ratio 2:8. How much water must be added to the mixture so that the ratio of water and milk becomes 3:8?

A. 75 ml **B. 65 ml** **C. 70 ml** **D. 60 ml**

23. A rice dealer bought 60 kg of rice worth Rs. 30 per kg and 40 kg of rice worth Rs. 35 per kg. He mixes the two and sells the mixture at Rs. 40 per kg. What is the percentage profit in this deal?

A. 18% B. 21% C. 23% D. 25%

24. A bottle of whisky contains 40% alcohol. If we replace a part of this whisky by another whisky containing 20% alcohol, the percentage of alcohol becomes 28%. What quantity of whisky is replaced?

A. 3/5 B. 3/4 C. 4/5 D. 2/5

25. In what ratio must wheat A at Rs. 10.50 per kg be mixed with wheat B at Rs. 12.30 per kg, so that the mixture be worth of Rs. 11 per kg?

A. 13:5 B. 18:3 C. 17:5 D. 11:5

26. In what ratio must a shopkeeper mix Peas and Soybean of Rs. 16 and Rs. 25 per kg respectively, so as to obtain a mixture of Rs. 19.50 ?

A. 9:5 B. 7:5 C. 11:7 D. 11:5

27. Sugar A worth Rs. 130/kg and B of Rs. 120/kg are mixed with a third variety C in the ratio of 1 : 1 : 2. If the mixture is worth Rs. 160, then find the price of third variety of sugar.

A. Rs. 195 B. Rs. 200 C. Rs. 225 D. Rs. 230

28. Two containers P and Q contain milk and water in the ratio of 5 : 2 and 7 : 6 respectively. Find the ratio in which these two mixtures can be mixed so that a new mixture formed in the container R is in the ratio of 8 : 5.

A. 5:6 B. 4:9 C. 7:9 D. 9:7

29. A jar full of whisky contains 40% alcohol. A part of this whisky is replaced by another containing 19% alcohol and now the percentage of alcohol was found to be 26%. The quantity of whisky replaced is:

A. 1/3 B. 2/3 C. 2/5 D. 3/5

30. The cost of Type 1 rice is Rs. 15 per kg and Type 2 rice is Rs. 20 per kg. If both Type 1 and Type 2 are mixed in the ratio of 2 : 3, then the price per kg of the mixed variety of rice is:

A. Rs. 18 B. Rs. 18.50 C. Rs. 19 D. Rs. 19.50

SERIES

1. 12, 21, 37, 73, 49, x.
   A) 92 B) 94 C) 86 D) 102 E) None of these

   ANS:- B) 94

2. 55, 56, 55, 59, 51, 60, x.
   A) 31 B) 33 C) 87 D) 57 E) None of these

   ANS:- B) 33

3. 60, 30, 70, 40, 80, 50, x.
   A) 70 B) 80 C) 90 D) 60 E) 95

   ANS:- C) 90

4. 2, 4, 12, 60, 420, x.
   A) 840 B) 1020 C) 3780 D) 4620 E) None of these

   ANS:- D) 4620

5. QPO, SRQ, UTS, WVU,
   A) AYZ B) ZYA C) YXW D) VWX E) XVZ

   ANS:- C) YXW

6. **EVIC, DWHB, CXGA,**

A) BYLO **B) BYMQ** **C) BYNO** **D) BYFZ**

E) CYQM

Ans:- D) BYFZ

7. **LN18, JQ25, HT32, FW39,**

A) DY43 **B) DZ46** **C) DA43** **D) DX43**

E) CZ46

Ans:-B) DZ46

8. **81, 80, x, 67, 51, 26.**

A) 74 **B) 78** **C) 79** **D) 76** **E) None of these**

Ans:- D) 76

9. **3, 6, 12, x, 33, 48**

A) 18 **B) 22** **C) 27** **D) 21** **E) None of these**

Ans:- D) 21

10. **………,ayw, gec, mki, sqo.**

A) bwz **B) zxw** **C) usq** **D) may** **E) xyv**

Ans:- C) usq

11. **HYD, IVF, KRH,……… , RGL.**

A) NMJ B) NMI C) MNJ D) MNI E) LMJ

Ans A) NMJ

12. **a_aab_b_cc**

A) abbcc B) abccb C) acbba D) ccbba

E) aabbcc

Ans A) abbcc

13. **a_a_bab_ab_baa_.**

A) baaab B) bbaab C) bbbab

D) aaaba E) aabba

Ans A) baaab

14. **MN_PPO_MM_OP_ON_.**

A) OOMNQ B) MNOPQ C) OMNPO

D) ONNPM E) PMNNO

Ans D) ONNPM

15. **24,51,105,213,427,861.**

A) 105 B) 213 C) 51

D) 427 E) 861

Ans D) 427

16. 5,7.5, 15, 37.5, 112.

A)15 B) 7.5 C) 37.5 D) 5 E) 112

Ans E) 112

17. 90, 72, 56,44,30

A) 90 B) 72 C) 56 D) 44 E) 30

Ans D) 44

18. BDE, FHI, JLM, NPQ, RSU

A) BDE B) FHI C) JLM D) NPQ E) RSU

Ans E) RSU

19. CEFH, JLMO, QSTU, XZAC

A) JLMO B) CEFH C) QSTU

D) XZAC E) None of these

Ans C) QSTU

20. E, G, K, M, O, S, W

A) G B) K C) O D) W E) S